

CENTRE FOR ADVANCED STUDIES

**DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH,
LUCKNOW**



EVALUATION SCHEME AND SYLLABI

For

**B.Tech. Ist Year
Mechatronics & Mechanical Engineering**

Effective from the Session: 2025-26

DR. A.P.J ABDUL KALAM TECHNICAL UNIVERSITY UTTAR PRADESH.
Sec-11, Jankipuram Vistar, Lucknow-226031, Uttar Pradesh, India

DETAILED SYLLABI

B.Tech. First Year

Effective from Session 2025-26

B.Tech. Mechatronics & Mechanical Engineering First Year, Semester-I

3-WEEKS STUDENT INDUCTION PROGRAMME

In the beginning of the session

										Evaluation Scheme			
SN	Subject Code	Subject Name	Type	Category	Period			Sessional Component		Sessional (SW)/(TS/PS)	End Semester Examination(ESE)	Total	Credit
					L	T	P	CT	TA	CT+TA	TE/PE	SW+ESE	Cr
1.	CAS 101	Engineering Physics	T	BS	3	1	0	20	10	30	70	100	4
2.	CAS 102	Chemistry for Engineers	T	BS	3	1	0	20	10	30	70	100	4
3.	CAS 103	Engineering Mathematics-I	T	BS	3	0	0	20	10	30	70	100	3
4.	CAS 105	English for Technical Writing	T	BS	3	1	0	20	10	30	70	100	4
5.	CME 101	Fundamentals of Mechanical Engineering	T	ES	3	0	0	20	10	30	70	100	3
6.	CAS 151	Engineering Physics Lab	P	BS	0	0	3	-	50	50	50	100	1
7.	CAS 152	Engineering Chemistry Lab	P	BS	0	0	3	-	50	50	50	100	1
8.	CME 153	Sensors & Actuators Lab	P	ES	0	0	3	-	50	50	50	100	1
9.	CAS 153	English Language Lab	P	ES	0	0	3	-	50	50	50	100	1
					15	3	12	-		350	550	900	22
10.	Vocational Training / Skill based courses (Mandatory for Exit after I year)				-	-	-	-	-	-	-	-	2

Abbreviation Used:

BS: Basic Science Course

ES: Engineering Science Course

HS: Humanities and Social Science Course

VA: Value Added Course

B.Tech. Mechatronics & Mechanical Engineering First Year, Semester-II

									Evaluation Scheme				
SN	Subject Code	Subject Name	Type	Category	Period			Sessional Component		Sessional (SW)(TS/PS)	End Semester Examination (ESE)	Total	Credit
					L	T	P	CT	TA	CT+TA	TE/PE	SW+ESE	Cr
1.	CAS 203	Engineering Mathematics-II	T	BS	3	0	0	20	10	30	70	100	3
2.	CME 201	Sensors and Actuators	T	ES	3	1	0	20	10	30	70	100	4
3.	CEE 202	Electrical & Electronics Engineering	T	ES	3	1	0	20	10	30	70	100	4
4.	CCS 202	Programming Language Fundamentals	T	ES	3	0	0	20	10	30	70	100	3
5.	CAS 204	Environment & Energy	T	HS	3	0	0	20	10	30	70	100	3
6.	CEE 252	Electrical & Electronics Engineering Lab	P	ES	0	0	3	-	50	50	50	100	1
7.	CCS 252	Programming for Problem Solving Lab	P	ES	0	0	3	-	50	50	50	100	1
8.	CME 252	Engineering Visualization & Design Lab	P	HS	0	0	3	-	50	50	50	100	1
9.	CME 251	IDEA Lab Workshop	P	ES	0	0	6	-	50	50	50	100	2
10.	CVA 252	Sports and Yoga/NSS	P	VA	0	0	3*		100	*100		*100	0
					15	2	15+3*			350+100*	550	900+100*	22
11.	Vocational Training / Skill based courses (Mandatory for Exit after I year)				-	-	-	-	-	-	-	-	2

*Compulsory Qualifying Audit Course

Abbreviation Used:

BS: Basic Science Course

ES: Engineering Science Course

HS: Humanities and Social Science Course

VA: Value Added Course

Summer Internship (4-week)/NPTEL Course(4 week) during summer break after Semester-II and same will be assessed/evaluated in the Semester-III

CAS 101: ENGINEERING PHYSICS

The objective of this course is to

- Understand the underlying mechanism involved in construction and working principles of various semiconductor devices.
- Study the fundamental concepts related to the dielectric, magnetic and energy materials.
- Identify the importance of nanoscale, quantum confinement and various fabrications techniques.
- Study the characteristics of lasers and optical fibres.
- Understand the basic principles of quantum physics and band theory of solids

Content	Contact Hours
Unit-1:Semiconductors and Devices	8
Intrinsic and extrinsic semiconductors – Hall effect - direct and indirect band gap semiconductors - construction, principle of operation and characteristics of P-N Junction diode, Zener diode and bipolar junction transistor (BJT)–LED, PIN diode, avalanche photo diode (APD) and solar cells, their structure, materials, working principle and characteristics.	
Unit-2:Dielectric, Magnetic and Energy Materials	8
Dielectric Materials: Basic definitions- types of polarizations (qualitative) - ferroelectric, piezoelectric, and pyroelectric materials – applications – liquid crystal displays (LCD) and crystal oscillators. Magnetic Materials: Hysteresis - soft and hard magnetic materials - magnetostriction, magnetoresistance - applications - bubble memory devices, magnetic field sensors and multiferroics. Energy Materials: Conductivity of liquid and solid electrolytes- superionic conductors - materials and electrolytes for super capacitors - rechargeable ion batteries, solid fuel cells	
Unit-3:Nanotechnology	8
Nanoscale, quantum confinement, surface to volume ratio, fabrication techniques of nanomaterials, characterization techniques - XRD, SEM &TEM - applications of nanomaterials.	
Unit-4:Lasers and Fiber Optics	8
Lasers: Laser beam characteristics-three quantum processes-Einstein coefficients and their relations- lasing action - pumping methods- ruby laser, He-Ne laser, CO2 laser, Argon ion Laser, Nd:YAG laser- semiconductor laser-applications of laser.	
Fiber Optics: Introduction to optical fiber- advantages of optical Fibers - total internal reflection- construction of optical fiber - acceptance angle - numerical	

aperture- classification of optical fibers- losses in optical fiber - optical fiber for communication system - applications.	
Unit-5: Fundamentals of Quantum Physics	8
Quantum Mechanics: Introduction to quantum physics, blackbody radiation – Stefan-Boltzmann’s law, Wein’s and Rayleigh-Jean’s law, Planck’s radiation law - photoelectric effect - Davisson and Germer experiment –Heisenberg uncertainty principle - Born interpretation of the wave function – time independent Schrodinger wave equation, particle in one-dimensional potential box.	

Course Outcomes:

Upon completion of the course, the student will be able to:

CO	CO Statement	Bloom’s Level
CO1	Understand the difference between conductor, semiconductor, and an insulator by classification of solids.	K2
CO2	Identify the role of semiconductor devices in science and engineering applications.	K2,K3
CO3	Explore the fundamental properties of dielectric, magnetic materials and energy for their applications.	K3
CO4	Appreciate the features and applications of Nanomaterials.	K2, K3
CO5	Understand various aspects of Lasers and Optical fiber and their applications in diverse fields. Understand physical world from fundamental point of view by the concepts of Quantum physics	K2

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Text Books and Reference Books:

1. Introduction to Solid State Physics, Charles Kittel, Wiley Eastern, 2019
2. Semiconductor Physics and Devices- Basic Principle – Donald A, Neamen, Mc Graw Hill, 4th Edition, 2021.
3. Quantum Physics, H.C. Verma, TBS Publication, 2nd Edition 2012.
4. M. E. Browne- Schaum’s Outline of Physics for Engineering and Science, Mc Graw Hill, Fourth Edition.
5. M. N. Avadhanulu, P.G. Kshirsagar & TVS Arun Murthy” A Text book of Engineering Physics”- S. Chand Publications, 11th Edition 2019.
6. Engineering Physics by Shatendra Sharma and Jyotsna Sharma, Pearson Publication, 2019
7. K. Pandey and S. Chaturvedi, Engineering Physics, Cengage Learning, 2nd Edition, 2022.
8. Essentials of Nanoscience & Nanotechnology by Narasimha Reddy Katta, Typical Creatives NANO DIGEST, 1st Edition, 2021.
9. Fundamentals of Physics – Halliday, Resnick and Walker, John Wiley & Sons, 11th Edition, 2018.
10. A.K. Bhandhopadhyaya - Nano Materials, New Age International, 1st Edition, 2007.
11. <https://onlinecourses.nptel.ac.in/>

CAS 102: CHEMISTRY FOR ENGINEERS

Course Objectives:

1. To enable the students to understand about the Chemistry of Atomic and Molecular structure, Chemistry of advanced Materials like Liquid crystals, Nanomaterials, Graphite & fullerenes and Green Chemistry.
2. To enable the students to understand and apply the detailed concepts of spectroscopic techniques and stereochemistry to identify the compounds, element etc.
3. To enable the students to understand and apply the concepts related to Electrochemistry, Batteries, Corrosion and Chemistry of Engineering Materials like cement.
4. To enable the students to understand and apply detailed concepts of water source, water impurities, hardness of water and boiler troubles used in industry as well as analysis of coal & determination of calorific values.
5. To enable the students to understand detailed concepts related to polymers, Polymerization, Polymer Blends and Polymer Composites.

Syllabus

Content	ContactHours
Unit-1: Atomic and Molecular Structure: Molecular orbital's of diatomic molecules, bond order, Magnetic characters and numerical problems. Chemistry of Advanced Materials: Liquid Crystals; Introduction, Types and Applications of liquid crystals, Industrially important materials used as liquid crystals. Nanomaterials; Introduction, preparation, characteristics and applications of nanomaterials such as carbon nanotubes (CNT), graphene, fullerenes etc. Introduction to two-dimensional nanomaterials. Green Chemistry: Introduction to green synthesis, synthesis of nanomaterials by green route.	8
Unit-2: Spectroscopic Techniques and Applications: Elementary idea and simple applications of UV, IR and NMR, Numerical problems. Stereochemistry: Structural isomers and stereoisomers, configurations and symmetry and chirality, enantiomers, diastereomers, optical activity, absolute configurations and conformational analysis. Isomerism in transitional metal compounds	8
Unit-3: Electrochemistry and Batteries: Basic concepts of electrochemistry. Batteries: Classification and applications of Primary Cells (Dry Cell) and Secondary Cells (Lead Acid battery). Corrosion: Introduction to corrosion, Types of corrosion, Cause of	8

corrosion, Corrosion prevention and control, Corrosion issues in specific industries (Power generation, Chemical processing industry, Oil & gas industry and Pulp & paper industries). Chemistry of Engineering Materials: Cement: Constituents, manufacturing, hardening and setting, deterioration of cement, Plaster of Paris (POP).	
Unit-4: Water Technology: Sources and impurities of water, Hardness of water, Boiler troubles, Techniques for water softening, Determination of Hardness and alkalinity, Numerical problems. Definition, classification & characteristics of a good fuel, Calorific Value, Determination of calorific value by Bomb Calorimeter, Fuel Cells introduction. Basics of thermodynamics.	8
Unit-5: Polymer Chemistry: Classification, techniques of polymerization, polymer blends and composites, conducting and biodegradable polymers, industrial applications of some commonly used plastics and rubbers.	8

Upon completion of the course, the student will be able to:

	Course Outcomes	Bloom's Level
CO-1	Get an understanding of the theoretical principles of chemistry of molecular structure, bonding and properties, Chemistry of advanced materials (liquid crystals, nanomaterials, graphite & fullerene) as well as the Principles of Green Chemistry.	K3
CO-2	Apply the fundamental concepts of determination of structure with various spectral techniques and stereochemistry.	K4
CO-3	Utilize the theory of construction of electrodes, batteries and fuel cells in redesigning new engineering products and categorize the reasons for corrosion and study methods to control corrosion and develop understanding of Chemistry of Engineering materials (Cement)..	K3
CO-4	Develop understanding of the sources, impurities and hardness of water, understand the concepts of thermodynamics	K2
CO-5	Develop the understanding of chemical structure of polymers and its effect on their various properties when used as engineering materials.	K3

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

TEXTBOOKS AND REFERENCE BOOKS:

1. Engineering Chemistry by Rath & Singh, 2nd Edition, Cengage Learning India Pvt Ltd Delhi.

2. Engineering Chemistry by S S Dara, S Chand & Co Ltd
3. Engineering Chemistry by Jain & Jain, S. Chand & Comp, New Delhi
4. Engineering Chemistry by K. Sesha Maheswaramma, Pearson
5. Engineering Chemistry by OG Palanna, McGraw Hill Education, New Delhi
6. Engineering Chemistry by Shashi Chawala, Dhanpat Rai Publishing Comp, New Delhi
7. University General Chemistry by CNR Rao, Laxmi Publications
8. <https://onlinecourses.nptel.ac.in/>

CAS 103: ENGINEERING MATHEMATICS-I

Course Objectives:

The objective of this course is to familiarize the graduate engineers with techniques in matrix, calculus, multivariate analysis and vector calculus. It aims to equip the students with standard concepts and tools from intermediate to advanced level that will enable them to tackle more advanced level of mathematics and applications that they would find useful in their disciplines.

The students will learn:

- The essential tools of matrices, Eigen values and its application in a Comprehensive manner.
- To apply the knowledge of differential calculus in the field of engineering.
- To deal with functions of several variables that is essential in optimizing the results of real-life problems.
- To apply integral calculus in various field of engineering and have a basic understanding of Beta and Gamma functions and application of Dirichlet's integral.
- To deal with vector calculus that is required in different branches of Engineering to graduate engineer.

Content	Contact Hours
Unit-1: Matrices	8
Elementary transformations, Inverse of a matrix, Rank of matrix, Solution of system of linear equations, Characteristic equation, Cayley-Hamilton Theorem and its application, Linear Dependence and Independence of vectors, Eigen values and Eigen vectors, Complex Matrices, Hermitian, Skew-Hermitian and Unitary Matrices, Applications to Engineering problems.	
Unit-2: Differential Calculus- I	8
Successive Differentiation (nth order derivatives), Leibnitz theorem, Curve tracing, Partial derivatives, Euler's Theorem for homogeneous functions, Total derivative, Change of variables.	
Unit-3: Differential Calculus-II	8
Expansion of functions by Taylor's and Maclaurin's theorems for functions of one and two variables, Maxima and Minima of functions of several variables, Lagrange's method of multipliers, Jacobians, Approximation of errors.	
Unit-4: Multiple Integration	8
Double integral, Triple integral, Change of order of integration, Change of variables, Beta and Gama function and their properties, Dirichlet's integral and its applications to area and volume, Liouville's extensions of Dirichlet's integral.	
Unit-5: Vector Calculus	8
Vector Differentiation: Gradient, Curl and Divergence and their Physical interpretation, Directional derivatives. Vector Integration: Line integral, Surface integral, Volume integral, Gauss's Divergence theorem, Green's theorem and Stoke's theorem (without proof) and their applications.	

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Understand the concept of complex matrices, Eigen values, Eigen vectors and apply the concept of rank to evaluate linear simultaneous equations	K2, K5
CO 2	Remember the concept of differentiation to find successive differentiation, Leibnitz Theorem, and create curve tracing, and find partial and total derivatives	K1, K6 & K5
CO 3	Applying the concept of partial differentiation to evaluate extrema, series expansion, error approximation of functions and Jacobians	K3, K5
CO 4	Remember the concept of Beta and Gamma function; analyze area and volume and Dirichlet's theorem in multiple integral	K1, K4
CO 5	Apply the concept of Vector Calculus to analyze and evaluate directional derivative, line, surface and volume integrals.	K3, K4 & K5

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Text Books:

1. B. V. Ramana, Higher Engineering Mathematics, McGraw-Hill Publishing Company Ltd., 2008.
2. B. S. Grewal, Higher Engineering Mathematics, Khanna Publisher, 2005.
3. R K. Jain & S R K. Iyenger, Advance Engineering Mathematics, Narosa Publishing House 2002.

Reference Books:

1. E. Kreyszig, Advance Engineering Mathematics, John Wiley & Sons, 2005.
2. Peter V. O'Neil, Advance Engineering Mathematics, Thomson (Cengage) Learning, 2007.
3. Maurice D. Weir, Joel Hass, Frank R. Giordano, Thomas, Calculus, Eleventh Edition, Pearson.
4. D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/Cole, 2005.
5. Veerarajan T., Engineering Mathematics for first year, McGraw-Hill, New Delhi, 2008.
6. Ray Wylie C and Louis C Barret, Advanced Engineering Mathematics, McGraw-Hill; Sixth Edition.
7. P. Sivaramakrishna Das and C. Vijayakumari, Engineering Mathematics, 1st Edition, Pearson Education.
8. Advanced Engineering Mathematics. Chandrika Prasad, Reena Garg, 2018.
9. H.K. Dass, Advanced Engineering Mathematics, H.K.Dass, S. Chand

CAS 203: ENGINEERING MATHEMATICS-II

Course Objectives:

The objective of this course is to familiarize the prospective engineers with techniques in ordinary differential equations, Laplace transform, sequence and series, Fourier series and complex variables. It aims to equip the students to deal with advanced level of mathematics and applications that would be essential for their disciplines.

The students will learn:

- The effective mathematical tools for the solutions of differential equations that model physical processes.
- The basic knowledge of Laplace transforms and its applications in solving differential equations.
- The tool for convergence of series and expansion of function using Fourier series for learning advanced Engineering Mathematics.
- The tools of differentiation of functions of complex variables that are used in various techniques dealing with engineering problems.
- The tools of integration of functions of complex variables that are used in various techniques dealing with engineering problems.

Content	Contact Hours
Unit-1: Ordinary Differential Equation of Higher Order	8
Linear differential equation of nth order with constant coefficients, Simultaneous linear differential equations, Second order linear differential equations with variable coefficients, Solution by changing independent variable, Method of variation of parameters, Cauchy-Euler equation, Application of differential equations in solving engineering problems.	
Unit-2: Laplace Transform	8
Laplace transform, Existence theorem, Properties of Laplace Transform, Laplace transform of derivatives and integrals, Unit step function, Laplace transform of periodic function, Inverse Laplace transform, Convolution theorem. Application of Laplace Transform to solve ordinary differential equations and simultaneous differential equations.	
Unit-3: Sequence and Series	8
Definition of Sequence and series with examples, Convergence of series, Tests for convergence of series, Ratio test, D' Alembert's test, Raabe's test, Comparison test. Fourier series, Half range Fourier sine and cosine series.	
Unit-4: Complex Variable–Differentiation	8
Functions of complex variable, Limit, Continuity and differentiability, Analytic functions, Cauchy- Riemann equations (Cartesian and Polar form), Harmonic function, Method to find Analytic functions, Milne's Thompson Method, Conformal mapping, Mobius transformation and their properties.	
Unit-5: Complex Variable –Integration	8

Complex integration, Cauchy- Integral theorem, Cauchy integral formula, Taylor's and Laurent's series, singularities and its classification, zeros of analytic functions, Residues, Cauchy's Residue theorem and its application.	
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Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Remember the concept differentiation to evaluate LDE of nth order with constant coefficient and LDE with variable coefficient of 2nd order.	K1 & K5
CO 2	Understand and apply the concept of Laplace Transform to evaluate differential equations.	K2 ,K3 & K5
CO 3	Understand the concept of convergence to analyze the convergence of series and expansion of the function for Fourier series.	K2 & K4
CO 4	Apply the concept of analyticity, Harmonic function and create the image of function applying conformal transformation.	K3, K6& K3
CO 5	Apply the concept of Cauchy Integral theorem, Cauchy Integral formula, singularity and calculus of residue to evaluate integrals.	K3& K5

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

Text Books:

1. B. V. Ramana, Higher Engineering Mathematics, Tata McGraw-Hill Publishing Company Ltd., 2008.
2. B. S. Grewal, Higher Engineering Mathematics, Khanna Publisher, 2005.
3. 3. R. K. Jain & S. R. K. Iyenger, Advance Engineering Mathematics ,Narosa Publishing - House, 2002.

Reference Books:

1. E. Kreyszig, Advance Engineering Mathematics, John Wiley & Sons, 2005.
2. Peter V. O'Neil, Advance Engineering Mathematics, Thomson (Cengage) Learning, 2007.
3. Maurice D. Weir, Joel Hass, Frank R. Giordano, Thomas, Calculus, Eleventh Edition, Pearson.
4. G.B Thomas, R L Finney, Calculus and Analytical Geometry, Ninth Edition Pearson, 2002.
5. James Ward Brown and Ruel V Churchill, Fourier Series and Boundary Value Problems, 8th Edition-McGraw-Hill
6. D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/Cole, 2005.
7. Veerarajan T., Engineering Mathematics for first year, McGraw-Hill, New Delhi, 2008.
8. Charles E Roberts Jr, Ordinary Differential Equations, Application, Model and Computing, CRC Press T&F Group.

9. Ray Wylie C and Louis C Barret, Advanced Engineering Mathematics, 6th Edition, McGraw-Hill.
10. James Ward Brown and Ruel V Churchill, Complex Variable and Applications, 8th Edition, McGraw-Hill.
11. P. Sivaramakrishna Das and C. Vijayakumari, Engineering Mathematics, 1st Edition, Pearson India Education Services Pvt. Ltd.
12. Advanced Engineering Mathematics By Chandrika Prasad, Reena Garg Khanna Publishing House, Delhi.
13. Laplace Transforms by Schaum's series, 2005 Edition, Spiegel Publication.
14. H.K. Dass, Advanced Engineering Mathematics, H.K.Dass, S. Chand

CEE 202: ELECTRICAL AND ELECTRONICS ENGINEERING

Course Objectives:

The students will learn the basic concepts of electrical and electronics engineering. They will study about various types of electronics devices and electrical machines along with their fundamental aspects. This will be useful to the students who wish to pursue higher studies in electrical or electronics or relevant fields.

Content	Contact Hours
Unit-1	8
PN Diode And Transistor: PN Diode Theory, PN Characteristic and Breakdown Region, PN Diode Application as Rectifier, Zener Diode Theory, Zener Voltage Regulator, Diode as Clamper and Clipper, Photodiode Theory, LED Theory, 7 Segment LED Circuit Diagram and Multi Colour LED, LASER Diode Theory and Applications, Bipolar Junction Transistor Theory, Transistor Symbols And Terminals, Common Collector, Emitter and Base Configurations, Different Biasing Techniques, Concept of Transistor Amplifier, Introduction to FET Transistor And Its Feature.	
Unit-2	8
Rectifiers: Half wave, center tapped and bridge full-wave, Zener diode regulator and voltage multiplier, clipping and clamping circuits. Transistors: Construction and characteristics of BJT, Transistor configuration: CB, CE, CC configuration, Transistor at low frequency, small signal low frequency transistor model (h-parameters), Analysis of transistor amplifier using h-parameters, Transistor biasing and bias stabilization: Operating point, Stability factor, Analysis of fixed bias, collector to base bias, Emitter resistance bias circuit and self-bias circuit, Bias compensation techniques.	
Unit-3	8
Voltage and current sources, dependent and independent sources, source conversion, DC circuit's analysis using mesh & nodal method, Thevenin's and superposition theorem, star-delta transformation. 1-phase AC circuits under sinusoidal steady-state, active, reactive, and apparent power, physical meaning of reactive power, power factor, 3-phase balanced and unbalanced supply, star and delta connections.	
Unit-4	8
Transformers: Magnetic Circuits: Review of laws of electromagnetism, Flux, MMF and their relation, analysis of the magnetic and electric circuit. Singlephase transformer: Basic concepts, constructional features, EMF equation, voltage, current, and impedance transformation, Equivalent circuits.	
Unit-5	8
Electrical Machines: DC Machines: Constructional features, working principle, emf equation, types of dc machines, and their characteristics. Induction Machines: Constructional features, working principle, emf	

equation, the concept of slip and torque–slip characteristics. Synchronous Machines: Constructional features, working principle and emf equation.	
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Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Understand basics of semiconductor theory and principle of diode operation.	K2
CO 2	To study the design and operation of rectifiers and transistor amplifiers.	K3
CO 3	To study and apply circuit theorems to AC and DC circuits.	K3
CO 4	Apply the concept of analyticity, Harmonic function and create the image of function applying conformal transformation.	K4
CO5	To understand the working principle of different machines	K2

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate, K6 – Create

TextBooks and Reference Books:

1. Ritu Sahdev, Basic Electrical Engineering, Khanna Book Publishing, 2022
2. I. J. Nagrath, D.P Kothari, Basic Electrical Engineering, Mc-Graw Hill Education, 2005.
3. Hayt and Kimberly, Engineering Circuit Analysis, Tata Mc-Graw Hill Publishing, 2013
4. <https://onlinecourses.nptel.ac.in/>

CAS 204: ENVIRONMENT AND ENERGY

Course Objectives:

1. Aims and objectives of environmental education emphasize the relationship between man and the environment and educate young people about the importance of nature and the environment.
2. Environmental education aims to impart ecological knowledge and promote environmentally conscious behavior towards nature.
3. It encourages young minds to take responsibility for protecting the natural environment protection through information and knowledge and to develop environmental awareness.
4. Incidentally, promoting awareness and a sense of respect for nature leads to a comprehensive understanding of the environment and a reasonable attitude towards protecting it.
5. The focus of environmental education is Awareness, Knowledge, Attitude, Skills, Capacity Building and Participation.

Syllabus

Content	Contact Hours
UNIT I	8
Environment: Definition, Types of Environment, Components of environment, Segments of environment, Scope and importance, Need for Public Awareness. Ecosystem: Definition, Types of ecosystem, Structure of ecosystem, Food Chain, Food Web, Ecological pyramid. Balance Ecosystem. Effects of Human Activities such as Food, Shelter, Housing, Agriculture, Industry, Mining, Transportation, Economic and Social security on Environment, Environmental Impact Assessment, Sustainable Development.	
UNIT II	8
Pollution and their Effects; Public Health Aspects of Environmental; Water Pollution, Air Pollution, Soil Pollution, Noise Pollution, Thermal Pollution, Radioactive Pollution, Marine Pollution and Solid waste management.	
UNIT III	8
Current Environmental Issues of Importance; Global Warming, Green House Effects, Climate Change, Acid Rain, Ozone Layer Formation and Depletion, Population Growth and Automobile pollution, Burning of paddy straw, concept of carbon credit, CO ₂ sequestration, environmental impact assessment	
UNIT IV	8
Basics of Energy: Classification of energy sources, Global and national energy scenario, Fossil and alternate fuels and its characterization. Energy Conservation Systems: General aspects of energy conservation and management. Energy conversion systems: Working principle, Basic components, General functioning and normal rating specifications of various energy conversion systems like Power plant, Pump, Refrigerator, Air-conditioner, Internal combustion engine, Solar PV cell, Solar water heating system, Biogas plant. Wind turbine, Fuel cells.	

UNIT V	8
Environmental Protection; Environmental Protection Act 1986, Initiatives by Non Governmental Organizations (NGO's), Human Population and the Environment: Population growth, Environmental Education, Women Education.	

Course Outcomes:

Upon completion of the course, the student will be able to:

	Course Outcomes	Bloom's Level
CO-1	Gain in-depth knowledge on natural processes that sustain life, and govern economy.	K2
CO-2	Differentiate between various types of environmental pollution along with their impacts and regulatory standards	K3
CO-3	Examine various global environmental issues and their management	K4
CO-4	Discuss the fundamental principles of energy, including classification, conservation and related policy frameworks and regulations. Get acquainted with the concept of energy systems and their components	K3
CO-5	Adopt sustainability as a practice in life, society and industry.	K3

TEXT BOOKS AND REFERENCE BOOKS:

1. Textbook of Environment and Ecology by Dave, Katewa & Singh, 2nd Edition, Cengage Learning India Pvt Ltd Delhi.
2. Daniel B. Botkin & Edward A Keller, Environmental Sciences, John Wiley & Sons.
3. R. Rajagopalan, Environmental Studies, Oxford University Press
4. Environmental Studies by S Deswal, Dhanpat Rai & Co.
5. <https://onlinecourses.nptel.ac.in/>

CCS 202: PROGRAMMING LANGUAGE FUNDAMENTALS

The objectives of this course are:

1. Understand Programming Basics: Develop a comprehensive understanding of fundamental programming concepts, including data types, control structures, and functions in C and Python.
2. Problem-Solving Skills: Apply logical thinking and algorithmic techniques to solve real-world problems using efficient code in both C/C++ and MATLAB.
3. Develop Code Efficiency: Write structured, modular, and optimized programs while adhering to best practices for code readability and maintainability.
4. Data Handling and File Operations: Explore techniques for handling data structures and performing file operations in C/C++ and MATLAB to manage data processing tasks.

Syllabus

Content	Contact Hours
Unit-1:	8
Introduction to Components of a Computer System: Memory, Processor, I/O Devices, Storage, Operating System, Concept of Assembler, Compiler, Interpreter, Loader and Linker. Idea of Algorithm: Representation of Algorithm, Flowchart, Pseudo Code with Examples, From Algorithms to Programs, Source Code. Programming Basics: Structure of C/C++/ MATLAB Program, Writing and Executing C/C++/MATLAB Program, Syntax and Logical Errors in Compilation, Object and Executable Code. Components of C/C++/MATLAB. Standard I/O in C/C++/MATLAB, Fundamental Data types, Variables and Memory Locations, Storage Classes.	
Unit-2:	8
Arithmetic Expressions and Precedence: Operators and Expression Using Numeric and Relational Operators, Mixed Operands, Type Conversion, Logical Operators, Bit Operations, Assignment Operator, Operator precedence and associativity. Conditional Branching: Applying if and Switch Statements, Nesting if and Else and Switch. Iteration and Loops: Use of While, do While and for Loops, Multiple Loop Variables, Use of Break, Goto and Continue Statements.	
Unit-3:	8
Arrays: Array Notation and Representation, Manipulating Array Elements, using Multi-Dimensional Arrays. Character Arrays and Strings, Structure, union, Enumerated Data types, Array of Structures, Passing Arrays to Functions. Functions: Introduction, Types of Functions, Functions with Array, Passing Parameters to Functions, Call by Value, Call by Reference, Recursive Functions.	
Unit-4:	8

Basic of searching and Sorting Algorithms: Searching & Sorting Algorithms (Linear Search, Binary search, Bubble Sort, Insertion and Selection Sort)	
Unit-5:	8
Pointers: Introduction, Declaration, Applications, Introduction to Dynamic Memory Allocation (Malloc, Calloc, Realloc, Free), String and String functions, Use of Pointers in Self-Referential Structures, Notion of Linked List (No Implementation) File Handling: File I/O Functions, Standard C Preprocessors, Defining and Calling Macros and Command-Line Arguments.	

Course Outcome:

At the end of course, the student will be able to

Course Outcome (CO)		Bloom's Level
CO1	To Develop Simple Algorithms for Arithmetic and Logical Problems.	K2,K3
CO2	To Translate the Algorithms to Programs & Execution (in C Language).	K3
CO3	To Implement Conditional Branching, Iteration and Recursion.	K3
CO4	To Decompose a Problem into Functions and Synthesize a Complete Program using Divide and Conquer Approach.	K4
CO5	To Use Arrays, Pointers and Structures to Develop Algorithms and Programs.	K2, K3

K1-Remember,K2-Understand,K3-Apply,K4-Analyze,K5-Evaluate,K6-Create

TextBooks and Reference Books:

1. Schaum's Outline of Programming with C by Byron Gottfried, McGraw-Hill
2. The C programming by Kernighan Brain W. and Ritchie Dennis M., Pearson Education .
3. Computer Basics and C Programming by V.Rajaraman, PHI Learning Pvt. Limited, 2015.
4. Computer Concepts and Programming in C, E Balaguruswami, McGraw-Hill
5. Computer Science A Structured Programming Approach Using C, by Behrouz A. Forouzan, Richard F. Gilberg, Thomson, Third Edition, Cengage Learning - 2007.
6. Let Us C By Yashwant P. Kanetkar.
7. Problem Solving and Program Design in C, by Jeri R.Hanly, Elliot B.Koffman, Pearson Addison- Wesley, 2006.
8. Programming in C by Kochan Stephen G. Pearson Education–2015.
9. Computer Concepts and Programming in C by D.S.Yadav and Rajeev Khanna, NewAge International Publication.
10. Computer Concepts and Programming by Anami, Angadi and Manvi, PHI Publication
11. Computer Concepts and Programming in C by Vikas Gupta, Wiley India Publication
12. Computer Fundamentals and Programming in C.Reema Thareja, OxfordPublication
13. <https://onlinecourses.nptel.ac.in/>

CME 201: SENSORS AND ACTUATORS

Course Objectives: At the end of this course, students should:

1. Apply contemporary knowledge to understand the classification of different types of measurement systems their static and dynamic characteristics.
2. Different types of sensors and actuators with their working principles for different domain of applications.
3. To provide skills to design a mechatronics system with the integration of sensors and actuators.

Syllabus

Content	Contact Hours
Unit-1:	8
Brief overview of measurement systems, classification, characteristics and calibration of different sensors.	
Unit-2:	8
Measurement of displacement, position, motion, force, torque, strain gauge, pressure flow, temperature sensor sensors, smart sensor. Optical encoder, tactile and proximity, ultrasonic transducers, opto-electrical sensor, gyroscope.	
Unit-3:	8
Principles and structures of modern micro sensors, micro-fabrication technologies: bulk micromachining, surface micromachining, LIGA, assembly and packaging.	
Unit-4:	8
Pneumatic and hydraulic systems: actuators, definition, example, types, selection. Pneumatic actuator. Electro-pneumatic actuator. Hydraulic actuator, control valves, valve sizing valve selection. Closed loop systems.	
Unit-5:	8
Electrical actuating systems: solid-state switches, solenoids, voice coil Electric motors; DC motors, AC motors, single phase motor; 3-phase motor; induction motor; synchronous motor; stepper motors. Piezoelectric actuator: characterization, operation, and fabrication; shape memory alloys. Stepper motors, servo motors.	

Course Outcome:

At the end of course, the student will be able to

Course Outcome (CO)		Bloom's Level
CO1	Understand the variety of measurement systems and their characteristics. Study of some basic sensors.	K2
CO2	Study the working principle and development of advanced sensors based on the different applied principles.	K3
CO3	Different micro-fabrication technologies for design and development of sensors and other MEMS devices.	K2

CO4	Study of different mechanical actuating system.	K2
CO5	Study of different electrical actuating system. Characterization, operation, and fabrication of piezoelectric actuator.	K2,K3

K1-Remember,K2-Understand,K3-Apply,K4-Analyze,K5-Evaluate,K6-Create

TextBooks and Reference Books:

1. John G. Webster, Editor-in-chief, "Measurement, Instrumentation, and Sensors Handbook", CRC Press (1999).
2. Jacob Fraden, "Handbook of modern Sensors", AIP Press, Woodbury (1997).
3. Nadim Maluf, "An Introduction to Microelectromechanical Systems Engineering", Artech House Publishers, Boston (2000).
4. Marc Madou, "Fundamentals of Microfabrication", CRC Press, Boca Raton (1997).
5. Gregory Kovacs, "Micromachined Transducers Sourcebook", McGraw-Hill, New York (1998).
6. E. O. Deobelin and D. Manik, "Measurement Systems – Application and Design", Tata McGraw-Hill (2004).
7. D. Patranabis, "Principles of Industrial Instrumentation", Tata McGraw-Hill, eleventh reprint (2004).
8. B. G. Liptak, "Instrument Engineers' Handbook: Process Measurement and Analysis", CRC (2003).
9. <https://onlinecourses.nptel.ac.in/>

CME 153: SENSORS AND ACTUATORS LAB

List of Experiments (indicative and not limited to)

1. Introduction about Sensor and its basic working principles.
2. Introduction to basic network theorems & circuits:
 - Norton theorem
 - Thevenin's theorem
 - Wheat stone bridge
 - KLC, KVL, and Maximum Power Transfer Theorem.
3. Introduction of Position, Proximity, Distance & Velocity Measuring Sensors. (Ultrasonic Sensor, IR Sensor, Radar Sensor, LVDT etc.)
4. Introduction of Different Touch and Pressure type Sensor.
5. Introduction of Force and Torque Measuring Sensor.
6. Introduction of Temperature Sensor. (To Measure the Characteristics of RTD, Thermocouple and LM35 Sensor)
7. Introduction of Flow Measuring Sensor.
8. Introduction of different rays, Waves and chemical Detecting Sensors. (Color, LDR, solar, etc.)
9. Introduction to different Actuators (Electrical, Pneumatic, Hydraulic). (SMA, DEA, Piezoelectric, motors)
10. DIY (Do It Yourself) project on sensors and actuators model/prototype design/development; using these develop a system contain least two sensors and two actuators.

CAS 105: ENGLISH FOR TECHNICAL WRITING

The objectives of this course are to enable students to:

1. To provide learning environment to practice listening, speaking, reading and writing skills.
2. To assist the students to carry on the tasks and activities through guided instructions and materials.
3. To effectively integrate English language learning with employability skills and training.
4. To provide hands-on experience through case-studies, mini-projects, group and individual presentations.

Content	Contact Hours
UNIT I: Vocabulary Building	8
The concept of Word Formation. Root words from foreign languages and their use in English. Acquaintance with prefixes and suffixes from foreign languages in English to form derivatives. Synonyms, antonyms, and standard abbreviations.	
UNIT II: Basic Writing Skills	8
Sentence Structures. Use of phrases and clauses in sentences. Importance of proper punctuation. Creating coherence. Organizing principles of paragraphs in documents. Techniques for writing precisely.	
UNIT III: Identifying Common Errors in Writing	8
Subject-verb agreement. Noun-pronoun agreement. Misplaced modifiers. Articles. Prepositions. Redundancies. Clichés	
UNIT IV: Nature and Style of Sensible Writing	8
Describing. Defining. Classifying. Providing examples or evidence. Writing introduction and conclusion.	
UNIT V: Writing Practices	8
Comprehension, Precis Writing, Essay Writing	

TEXTBOOK AND REFERENCES:

1. Communication Skills in English. Anjana Tiwari. Khanna Book Publishing. 2023.
2. Practical English Usage. Michael Swan. OUP. 1995.
3. Remedial English Grammar. F.T. Wood. Macmillan.2007
4. On Writing Well. William Zinsser. Harper Resource Book. 2001
5. Study Writing. Liz Hamp-Lyons and Ben Heasley. Cambridge University Press. 2006.
6. Communication Skills. Sanjay Kumar and PushpLata. Oxford University Press. 2011.
7. Exercises in Spoken English. Parts. I-III. CIEFL, Hyderabad. Oxford University Press
8. <https://onlinecourses.nptel.ac.in/>

Course Outcomes:

Upon completion of the course, the student will be able to:

	Course Outcomes
CO-1	Understand vocabulary building techniques
CO-2	Understand basic writing skills

CO-3	Apply the writing skills to identify common errors in writing
CO-4	Understand and practice the importance of writing a good introduction and conclusion
CO-5	Apply the writing skills to writing comprehension, précis and essay writing

CME 101: FUNDAMENTALS OF MECHANICAL ENGINEERING

Course Objectives: The objective of this course is to enable students to develop an understanding of the basics of mechanical engineering and be able to apply this fundamental knowledge towards the working of mechanical & mechatronics systems.

Syllabus

Content	Contact Hours
UNIT I: Introduction to Mechanics	8
Force moment and couple, principle of transmissibility, Varignon's theorem. Resultant of force system- concurrent and non-concurrent coplanar forces, Types of supports (Hinge, Roller) and loads (Point, UDL, UVL), free body diagram, equilibrium equations and Support Reactions.	
UNIT II: Mechanics of Solids	8
Normal and shear Stress, strain, Hooke's law, Poisson's ratio, elastic constants and their relationship, stress-strain diagram for ductile and brittle materials, factor of safety.	
UNIT III: Fundamentals of Mechanisms	8
Mechanisms: Definition – Machine and Structure – Kinematic link, pair and chain – classification of Kinematic pairs – Constraint & motion - Degrees of freedom, Slider crank mechanisms.	
UNIT IV: Overview of Transmission Drives	8
Power Screws: Forms of threads, multiple threads, Efficiency of square threads, Design of screw jack. Gearing and cams: Gear profile and geometry-nomenclature of spur and helical gears, law of gearing simple and compound gear trains. Cams & Followers, Cam-follower arrangements.	
UNIT V: Introduction to Mechatronics	8
Introduction to Mechatronic Systems: Evolution, scope, advantages and disadvantages of Mechatronics, Industrial applications of Mechatronics, Introduction to autotronics, bionics, and avionics and their applications. Sensors and Transducers: Types and their characteristics.	

Course Outcomes:

Upon completion of the course, the student will be able to:

	Course Outcomes	Bloom's Level
CO-1	Apply the concept of force resolution to solve basic problems	K3
CO-2	Understand the concept of stress-strain and elastic constants	K2
CO-3	Understand the fundamental of mechanisms and degree of freedom	K2
CO-4	Understand the working of transmission drives, gears and cams	K2
CO-5	Understand the working principle of different mechatronics systems with their advantages and disadvantages and applications	K2

TEXTBOOK AND REFERENCES:

1. Basic Mechanical Engineering, G Shanmugam, S Ravindran, McGraw Hill
2. Engineering Mechanics: Statics, Schaum Series, McGraw Hill
3. Basic Mechanical Engineering, M P Poonia and S C Sharma, Khanna Publishers
4. Mechatronics : Principles, Concepts and Applications, Nitaigour Mahalik, McGraw Hill
5. Mechatronics, As per AICTE: Integrated Mechanical Electronic Systems, K.P. Ramachandran,
6. G.K. Vijayaraghavan, M.S.Balasundaram, Wiley India
7. <https://onlinecourses.nptel.ac.in/>

CEE 252: ELECTRICAL AND ELECTRONICS ENGINEERING LAB

List of Experiments (indicative and not limited to)

1. To study the V-I characteristics of an incandescent lamp.
2. To measure single phase power by using three ammeter method.
3. To measure the single-phase power by using three voltmeter method.
4. To perform short circuit test on a single-phase transformer.
5. To perform open circuit test on a single-phase transformer.
6. To measure three phase power by using two wattmeter method.
7. To study the PN Junction diode characteristics.
8. To design the half wave and full wave rectifiers circuits.
9. To study CB, CE, CC input and output characteristics.

CAS 151: ENGINEERING PHYSICS LAB

Course Objective:

1. To develop the practical skill in Engineering Physics lab.
2. To gain hands on experience in handling of instruments, calibration of equipment, measure basic parameters etc.
3. To understand the principles of optics instruments along with underlying physics.
4. To explore the fluid mechanics and thermal properties
5. To analyze various electrical and electro-magnetic phenomena

Course Content:

List of Experiments:

1. Determination of moment of inertia of a fly wheel.
2. Measurement of angle of prism and the angle of minimum deviation for parallel rays on spectrometer and hence find the refractive index of the material of the prism.
3. To determine the radius of curvature of the convex surface of a lens by Newton's Rings.
4. Determination of the frequency of a tuning fork by Melde's apparatus.
5. Verification of Faraday and Lenz's law of induction.
6. Determination of earth's horizontal magnetic field intensity and magnetic moment of a magnet by employing magnetometer.
7. Determination of self-inductance of a coil with Anderson's Bridge.
8. Determination of
 - i) Plank's constant and work function using photo electric cell.
 - ii) Verification of inverse square law of radiation using a photo electric cell.
9. Measurement of unknown wavelength of laser with a diffraction grating.
10. Determination of unknown resistance by Carrey-Foster bridge method.
11. Determination of Young's Modulus of elasticity of the material of a bar by the method of flexure.
12. Determination of surface tension of a liquid by capillary tube method.
13. Determination of the co-efficient of viscosity of glycerin by Stoke's method.
14. Determination of rigidity modulus of the material of the wire by Torsional Pendulum.

Virtual Lab Experiments

1. Energy Band Gap of Semiconductor:

<https://bop-iitk.vlabs.ac.in/exp/energy-band-gap/simulation.html>

2. To study polarization of light using He-Ne Laser

<https://bop-iitk.vlabs.ac.in/exp/he-ne-laser-polarization/simulation.html>

3. To measure the specific rotation of cane sugar

<https://bop-iitk.vlabs.ac.in/exp/cane-sugar-rotation/simulation.html>

4. Measurement of the wavelength of monochromatic source of light with the help of Fresnel's Bi prism

<https://bop-iitk.vlabs.ac.in/exp/fresnel-biprism/simulation.html>

5. Measurement of Numerical aperture of optical fibre

<https://bop-iitk.vlabs.ac.in/exp/numerical-aperture-measurement/simulation.html>

6. Verification of Stefan's law

<https://bop-iitk.vlabs.ac.in/exp/stefans-law/simulation.html>

Course Outcome:

CO-1	Apply fundamental principles of mechanics and optics to experimentally determine physical properties such as the moment of inertia, refractive index, and radius of curvature of lenses.
CO-2	Analyze wave phenomena and electromagnetic principles by measuring frequency using Melde's apparatus, verifying Faraday's and Lenz's laws, and determining the Earth's magnetic field intensity.
CO-3	Evaluate electrical and quantum mechanical properties through experiments on self-inductance, Planck's constant, work function, and the inverse square law of radiation.
CO-4	Demonstrate the ability to measure material properties such as Young's modulus, surface tension, viscosity, and rigidity modulus using appropriate experimental methods.
CO-5	Develop experimental skills and data analysis techniques by using precision instruments like spectrometers, magnetometers, and diffraction gratings to investigate fundamental physical laws.

CAS 152: ENGINEERING CHEMISTRY LAB

The objective of this lab is to enable students to:

1. Gain knowledge to the basic tools, instruments and techniques used in chemistry
2. Attain a understanding of the theoretical concepts with the help of experiments
3. Gain hands on experience in handling instruments and using them
4. Understand the mechanism involved in the synthesis and reactions
5. Analyze the outcome of the experiments

LIST OF EXPERIMENTS (Any 10 experiments)

1. Calibration of Analytical Equipment and apparatus.
2. Determination of Hardness of water sample by EDTA method.
3. Determination of rate constant of a reaction.
4. Determination of chloride content of water
5. Determination of pH by titrimetric method.
6. Determination of surface tension of given liquid.
7. Synthesis of nanomaterial by bottom up method
8. Determination of viscosity of a given liquid by viscometer.
9. Acid Base titration
10. Determination of available chlorine in bleaching powder.
11. Preparation of phenol formaldehyde (PF) or urea formaldehyde (UF) resin.
12. Determination of unknown compounds using spectroscopic techniques
13. Preparation of thermosetting/ thermoplastic polymer composite
14. Determination of cell constant and conductance of a solution.
15. Element detection and identification of functional groups in organic compounds.
16. Saponification/acid value of an oil

Experiments may also be done through Virtual Lab mode, some of which are given here for reference:

S. No.	Experiment Name	Experiment Link(s)
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1	Determination of surface tension and viscosity	<u>Virtual Labs</u>
2	Determination of rate constant of a reaction	<u>Virtual Labs</u>
3	Determination of cell constant and conductance of solutions	<u>Virtual Labs</u>
4	Demonstration of the decrease in surface tension of water (including hard water) by the addition of soaps and detergents	<u>Virtual Labs</u>
5	Determination of amount of substance in a solution of unknown concentration using titrimetric methods	<u>Acid Base Titration (Simulator) : Inorganic Chemistry Virtual Lab : Chemical Sciences : Amrita Vishwa Vidyapeetham Virtual Lab</u>
6	Identification of unknown compounds using spectroscopic techniques	<u>Virtual Labs</u>

Course Outcomes:

Upon completion of the course, the student will be able to:

	Course Outcomes	Bloom's Level
CO-1	Get an understanding of the use of different analytical instruments.	K3
CO-2	Measure the molecular/system properties such as surface tension, viscosity, conductance of solution, chloride and iron content in the water.	K3
CO-3	Measure the hardness and alkalinity of the water	K3
CO-4	Know the fundamental concepts of the preparation of phenol formaldehyde & urea formaldehyde resin, adipic acid and Paracetamol.	K3
CO-5	Estimate the rate constant of reaction.	K3

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create

CCS 252: PROGRAMMING FOR PROBLEM SOLVING LAB

Programming for problem solving lab includes coding, experiments in C/C++/MATLAB (indicative and not limited to)

1. WAP that accepts the marks of 5 subjects and finds the sum and percentage marks obtained by the student.
2. WAP that calculates the Simple Interest and Compound Interest. The Principal, Amount, Rate of Interest and Time are entered through the keyboard.
3. WAP to calculate the area and circumference of a circle.
4. WAP that accepts the temperature in Centigrade and converts into Fahrenheit using the formula $C/5 = (F-32)/9$.
5. WAP that swaps values of two variables using a third variable.
6. WAP that checks whether the two numbers entered by the user are equal or not.
7. WAP to find the greatest of three numbers.
8. WAP that finds whether a given number is even or odd.
9. WAP that tells whether a given year is a leap year or not.
10. WAP that accepts marks of five subjects and finds percentage and prints grades according to the following criteria:
 - Between 90-100% ---- Print 'A'
 - 80-90% ----- Print 'B'
 - 60-80% ----- Print 'C'
 - Below 60%-----Print 'D'
11. WAP that takes two operands and one operator from the user, perform the operation, and prints the result by using Switch statement.
12. WAP to print the sum of all numbers up to a given number.
13. WAP to find the factorial of a given number.
14. WAP to print sum of even and odd numbers from 1 to N numbers.
15. WAP to print the Fibonacci series.
16. WAP to check whether the entered number is prime or not.
17. WAP to find the sum of digits of the entered number.
18. WAP to find the reverse of a number.
19. WAP to print Armstrong numbers from 1 to 100.
20. WAP to convert binary number into decimal number and vice versa.
21. WAP that simply takes elements of the array from the user and finds the sum of these elements.
22. WAP that inputs two arrays and saves sum of corresponding elements of these arrays in a third array and prints them.
23. WAP to find the minimum and maximum element of the array.
24. WAP to search an element in an array using Linear Search.
25. WAP to sort the elements of the array in ascending order using Bubble Sort technique.
26. WAP to add and multiply two matrices of order nxn.
27. WAP that finds the sum of diagonal elements of a mxn matrix.
28. WAP to implement strlen (), strcat (), strcpy () using the concept of Functions.
29. Define a structure data type TRAIN_INFO. The type contain Train No.: integer type Train name: string Departure Time: aggregate type TIME Arrival Time: aggregate type TIME Start station: string End station: string

The structure type Time contains two integer members: hour and minute.
Maintain a train timetable and implement the following operations:

- a. List all the trains (sorted according to train number) that depart from a particular section.
 - b. List all the trains that depart from a particular station at a particular time.
 - c. List all the trains that depart from a particular station within the next one hour of a given time.
 - d. List all the trains between a pair of start station and end station.
30. WAP to swap two elements using the concept of pointers.
 31. WAP to compare the contents of two files and determine whether they are same or not.
 32. WAP to check whether a given word exists in a file or not. If yes then find the number of times it occurs.

Note:

- a) *The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner*
- b) *The subject teachers are suggested to use the concept of project based learning. The subject teacher may give certain use cases/case studies where student is able to apply multiple concepts in one single program*

Mapping with Virtual Lab

Name of the Lab	Name of the Experiment
Problem Solving Lab	Numerical Representation
	Beauty of Numbers
	More on Numbers
	Factorials
	String Operations
	Recursion
	Advanced Arithmetic
	Searching and Sorting
	Permutation
	Sequences

Course Outcomes:

At the end of course, the student will be able to:

Course Outcome		Bloom's Level
CO 1	Able to implement the algorithms and draw flowcharts for solving Mathematical and Engineering problems.	K3, K4
CO 2	Demonstrate an understanding of computer programming language concepts.	K3, K2
CO 3	Ability to design and develop Computer programs, analyzes, and interprets the concept of pointers, declarations, initialization, operations on pointers and their usage.	K6, K4
CO 4	Able to define data types and use them in simple data processing applications he/she must be able to use the concept of array of structures.	K1, K5

CO 5	Develop confidence for self-education and ability for life-long learning needed for Computer language.	K3, K4
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K1- Remember, K2- Understand, K3- Apply, K4- Analyze , K5- Evaluate , K6- Create

CAS 153: ENGLISH LANGUAGE LAB

Course Objectives:

1. To facilitate software-based learning to provide the required English Language proficiency to students.
2. To acquaint students with specific dimensions of communication skills i.e. Reading, Writing, Listening, Thinking and Speaking.
3. To train students to use the correct and error-free writing by being well versed in rules of English grammar.
4. To cultivate relevant technical style of communication and presentation at their work place and also for academic uses.
5. To enable students to apply it for practical and oral presentation purposes by being honed up in presentation skills and voice-dynamics.

Professional Communication Lab shall have two parts:

1- Interactive Communication Skills:

Students should practice the language with variety of activities and exercises based on employability skills. Interactive and communicative practical with emphasis on oral presentation/spoken communication, based on International Phonetic Alphabets (I.P.A.)

LIST OF PRACTICALS

1. Group Discussion: Practical based on Accurate and Current Grammatical Patterns.
2. Conversational Skills for Interviews under suitable Professional Communication Lab conditions with emphasis on Kinesics.
3. Communication Skills for Seminars/Conferences/Workshops with emphasis on Paralinguistic/Kinesics.
4. Presentation Skills for Technical Paper/Project Reports/proposals based on proper Stress and Intonation Mechanics
5. Official/Public Speaking practice sessions based on suitable Rhythmic Patterns.
6. Theme Presentation/Keynote Presentation based on correct methodologies of argumentation
7. Individual Speech Delivery/Conferencing with skills to defend Interjections/Quizzes.
8. Argumentative Skills/Role Play Presentation with Stress and Intonation.
9. Comprehension Skills based on Reading and Listening Practical's on a model Audio
10. Startup presentations, Video portfolio, Extempore, Roleplay, Just A Minute (JAM) etc.

2- Computer assisted software based Language Learning:

Software based self guided learning to provide the required English language proficiency to students from an employability and career readiness stand point. The software should align to Common European Framework of Reference for Languages (CEFR) and deliver a CEFR level-B2 upon completion

Course Outcomes:

Upon completion of the course, the student will be able to:

	Course Outcomes
CO-1	Students will be able to understand the basic objective of the course by being acquainted with specific dimensions of communication skills i.e. Reading, Writing, Listening, Thinking and Speaking.
CO-2	Students would be able to create substantial base by the formation of strong professional vocabulary for its application at different platforms and through numerous modes as Comprehension, reading, writing and speaking etc.
CO-3	Students will apply it at their workplace for writing purposes such as Presentation/official drafting/administrative communication and use it for document/project/report/research paper writing.
CO-4	Students will be made to evaluate the correct and error-free writing by being well-versed in rules of English grammar and cultivate relevant technical style of communication & presentation at their workplace and also for academic uses.
CO-5	Students will apply it for practical and oral presentation purposes by being honed up in presentation skills and voice dynamics. They will apply techniques for developing interpersonal communication skills and positive attitude leading to their professional competence.

Suggested Softwares:

- *Oxford Achiever* by Oxford University Press.
- *Cambridge English Empower* by Cambridge University Press.
- *MePro*. By Pearson India Education Services Pvt .Ltd.
- *New Interactions* by McGraw-Hill India.

Reference Books:

1. Word Power Made Easy by Norman Lewis, W.R.Goyal Pub. & Distributors, 2009, Delhi.
2. Manual of Practical Communication by L.U.B. Pandey; A.I.T.B.S. Publications India Ltd.; KrishanNagar, 2013, Delhi
3. Practical Communication Process & Practice, L.U.B. Pandey: A.I.T.B.S. Pub. India Ltd Krishna Nagar, Delhi, 2013.
4. English Grammar and Usage by R.P.Sinha, Oxford University Press, 2005, New Delhi.
5. English Grammar, Composition and Usage by N.K.Agrawal&F.T.Wood, Macmillan India Ltd., NewDelhi.
6. Effective Communication Skill, Kulbhusan Kumar, R S Salaria, Khanna Publishing House
7. English Grammar & Composition by Wren & Martin, S.Chand&Co.Ltd., NewDelhi.
8. Communication Skills for Engineers and Scientists, Sangeeta Sharma et.al. PHI Learning Pvt. Ltd, 2011, NewDelhi.
9. Personality Development, Harold R. Wallace &L. Ann Masters, Cengage Learning, New Delhi
10. Personality Development & Soft Skills, Barun K.Mitra, Oxford University Press, 2012 New Delhi.
11. Business Correspondence and Report Writing by Prof. R.C.Sharma& Krishna Mohan, Tata Mc Graw Hill &Co.Ltd., 2001, New Delhi.
12. Developing Communication Skills by Krishna Mohan, Meera Bannerji-Macmillan India Ltd. 1990, Delhi.
13. SpokenEnglish A manual of Speech and Phonetics by R.K. Bansal

&J.B.Harrison, Orient Blackswan, 2013, NewDelhi.

14. Business English by KenTaylor, Orient Blackswan, 2011, NewDelhi

CME 252: ENGINEERING VISUALIZATION & DESIGN LAB

Course Objectives: The objective of this lab is to learn the basics of engineering drawing and implement the knowledge of this subject as an engineering design. The lab curriculum is designed for 1st year UG students for all engineering specializations. At the end of the course students will be able to implement their ideas in various applications and create job opportunities.

UNIT - I

Lines Lettering and Dimensioning: Types of lines, Lettering, Dimensioning, Geometrical Constructions, Polygons. Scales: Plain scales, Diagonal scales, Scale of chords.

UNIT - II

Curves used in Engineering Practice: Ellipse, Parabola, Hyperbola, normal and tangents to these curves, Involute, Cycloid, Epi-cycloid, Hypocycloid, Spiral, Helix on cone and cylinder.

UNIT- III

Orthographic projection of points: Principles of Orthographic projection, Projections of points. Projections of Lines: Projections of a line parallel to one of the reference planes and inclined to the other, line inclined to both the reference planes, Traces, Projections of Planes: Projections of a plane perpendicular to one of the reference planes and inclined to the other, Oblique planes.

UNIT - IV

Projections of Solids: Projections of solids whose axis is parallel to one of the reference planes and inclined to the other, axis inclined to both the planes. Section of Solids: Sectional planes, Sectional views - Prism, pyramid, cylinder and cone, true shape of the section.

UNIT - V

Isometric views: Isometric axis, Isometric Planes, Isometric View, Isometric projection, Isometric views – simple objects. Assembly drawings of the machine parts.

NOTE: Interpretation of drawings: Introduction of CAD package to construct a simple solid model, using a CAD package to construct solid models and generating orthographic, isometric, sectional views with dimensioning, Assembly of components and generation of corresponding drawings. Animation of machines in CAD.

Major Equipments:

Drawing sheet, Drawing board, Mini drafter, HB/1H/2H Pencils, Eraser, Protractor, scale, etc or using AutoCAD

List of Experiments

S. No. Description

1. Lettering, Dimensioning,
2. Plan Scale, Diagonal Scale, Scale of Chords
3. Geometrical construction of Engineering Curves
4. Introduction of Projection of Points and Lines
5. Traces of lines
6. Projection of Planes
7. Projections of Regular Solids

8. Sectional Views of Solids
9. Isometric views
10. Orthographic to isometric

Virtual Labs

1: Basics of Engineering Drawing

<https://eg-nitk.vlabs.ac.in/exp/engineering-drawing/simulation.html>

2: Projection of Points

<https://eg-nitk.vlabs.ac.in/exp/projections-of-points/simulation.html>

3: To draw projections of Line inclined with both VP and HP

<https://eg-nitk.vlabs.ac.in/exp/simple-positions/simulation.html>

4: To draw projections of Line inclined with both VP and HP and trace it

<https://eg-nitk.vlabs.ac.in/exp/projections-lines-inclined/simulation.html>

Text Books:

1. Engineering Graphics, N.D. Bhatt and V.M. Panchal, Charotar Publishers.
2. Jolhe, D. A., Engineering drawing, Tata McGraw Hill

Reference Books:

1. Engineering Drawing, Agarwal, B, McGraw Hill Education, 2015, second edition.
2. Shah, M. B. and Rana, B. C., Engineering Drawing, Pearson Education, 2009
3. K.V. Natarajan, A text book of Engineering Graphics, Dhanalakshmi Publishers, Chennai, 2006.
4. AutoCAD 2007 Bible E. Finkelstein, Wiley Publishing Inc.

CME 251: IDEA LAB WORKSHOP

SUGGESTIVE LIST OF EXPERIMENTS:

S. No.	Mechanical Workshop	Duration
1	Introduction to product design To study different tools used in SolidWorks. 2D and 3D part design in SolidWorks. Virtual experiments on PCB Design, Fabrication and Digital Fabrication of flexible circuit board etc.	3 Hours
2	Introduction to Mechanical workshop material, tools and machines To study layout, safety measures and different engineering materials (mild steel, medium carbon steel, high carbon steel, high speed steel and cast iron etc.) used in workshop. To study and use of different types of tools, equipments, devices & machines used in fitting, sheet metal and welding section. To determine the least count of vernier caliper, vernier height gauge, micrometer (Screw gauge) and take different reading over given metallic pieces using these instruments.	3 Hours
3	Machine shop Demonstration of working, construction and accessories for Lathe machine. Perform operations on Lathe - Facing, Plane Turning, step turning, taper turning, threading, knurling and parting. Virtual machining simulation using CAD/CAM tools involves creating a design model, creating workpiece, choosing manufacturing set up, defining machining sequences, generating toolpath, checking the results, and post-processing etc. Generate machining toolpath for complex 3D Shapes suitable for machining on the CNC minimal machine. Simulate the generated toolpath on CNC simulator and estimate machining time.	3 Hours
4	Fitting shop To study about different hand tools used in fitting shop. Preparation of U or V -Shape Male Female Work piece which contains: Filing, Sawing, Drilling, Grinding.	3 Hours
5	Welding Shop Introduction to BI standards and reading of welding drawings. Practice of making following operations Butt Joint, Lap Joint or T Joint by TIG Welding/MIG Welding/Arc Welding Train students with cutting-edge welding technologies on virtual welding simulators.	3 Hours
6	Moulding and Casting Shop Introduction to Patterns, pattern allowances, ingredients of moulding	3 Hours

sand and melting furnaces.

To study the different tools used in Foundry shop.

Demo of mould preparation and Aluminium casting Practice – Study and Preparation of Plastic mould

Virtual experiments on Molding and Casting of parts and components

7

CNC Shop

Study of main features and working parts of CNC machine and accessories that can be used. 3 Hours

Perform different operations on metal components using any CNC machines

8

Sheet metal shop

To study different types of Hand tools used in Sheet metal shop. 3 Hours

To prepare a square tray of given dimensions using a Galvanized iron (G.I) sheet.

To prepare a Funnel of given dimensions using a G.I. sheet.

9

To prepare a product using 3D printing

To study the different types of 3D printing processes. 3 Hours

Design and development of 3D printed part or component on any available 3D printer.

Virtual experiments on different types 3D scanning of objects.

Virtual experiments on different types of 3D Printing processes including their construction, process capabilities, material selection, parameter selection, post processing and defect analysis etc.

The students will be able to		Blooms Taxonomy
CO1	Use various engineering materials, tools, machines and measuring equipments.	K3
CO2	Perform machine operations in lathe and CNC machine.	K3
CO3	Perform manufacturing operations on components in fittings shop.	K3
CO4	Perform operations in welding, moulding, casting and gas cutting.	K3
CO5	Fabricate a job by 3D printing manufacturing technique	K3

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create

Text Books:

1. Basic Workshop Technology: Manufacturing Process, Felix W.; Independently Published, 2019.
2. Workshop Processes, Practices and Materials; Bruce J. Black, Routledge publishers, 5th Edn. 2015.
3. A Course in Workshop Technology Vol I. & II, B.S. Raghuwanshi, Dhanpath Rai & Co., 2015 & 2017.

Reference Books:

1. Workshop Practice by H. S. Bawa, Tata-McGraw Hill, 2004
2. Mechanical Workshop Practice, K C John, PHI
3. Workshop Practice Vol 1, and Vol 2, by Hazra Choudhary, Media promoters and Publications

4. CNC Fundamentals and Programming, By P. M. Agrawal, V. J. Patel, Charotar Publication.
5. Technology of machine tools, Steve F. Krar, Arthur R. Gill and Peter Smid, McGraw Hill Education (India) Pt. Ltd., 2013.
6. Engineering Practices Laboratory Manual, Ramesh Babu.V., VRB Publishers Private Limited, Chennai, Revised edition, 2013 – 2014.
7. Engineering Practices Lab Manual; T.Jeyapoovan, Vikas Pub, 4th Edn.2008.
8. Mechanical Workshop Practice; John K.C., PHI, 2010.
9. Wiring Estimating, Costing and Contracting; Soni P.M. & Upadhyay P.A.; Atul Prakashan, 2021-22.
10. H Gerling, All about Machine Tools, New Age International, 1995.
11. W A J Chapman, Workshop Technology, Oxford IBH, 1975.

Online Resources:

1. Different Trade E-Books (Fitting, Plumbing, Welding, Carpentry, Foundryman, Turner and House Wiring etc.) developed by National Instructional Media Institute, Chennai. Directorate General of Training, Ministry of Skill Development & Entrepreneurship, Govt.of India. (<https://bharatskills.gov.in>).
2. 3D Printing Virtual Simulation Lab (<https://3dp-dei.vlabs.ac.in/Introduction.html>)
3. Fabrication Lab (<https://fab-coep.vlabs.ac.in/Introduction.html>)
4. Mechanisms and Robotics Lab (<https://mr-iitkgp.vlabs.ac.in/Introduction.html>)
5. Micromachining lab (<https://mm-coep.vlabs.ac.in/Introduction.html>)